

Citation-grounded Medical Record Summarization PoC

Evidence-first AI draft summary workflow with clinician review, citation validation, audit trail and proxy evaluation

Clinician-review-only draft output. Uses mock / de-identified / proxy data. Not a claim of hospital deployment, live EHR validation, or autonomous diagnosis, treatment or prescribing.

2026 • 06 • 24

PREPARED FOR Vinmec leadership review

The bridge from PoC evidence to a governed Vinmec pilot



Problem

Records are fragmented and slow to review. Generic LLM summaries can omit or invent clinically important details with no visible source evidence.



Solution

A citation-grounded RAG workflow drafts a summary scoped to patient/encounter evidence, then requires clinician edit, approve or reject before any final lock.



Evidence

Doctor review workflow, citation/source view, unsupported-claim visibility, audit-oriented flow, Admin Evaluation dashboards and a verified Docker Compose local staging demo.



Boundary & next step

Results are proxy evaluation only — no live EHR validation, no hospital deployment, no autonomous decision authority. Next step: design a Vinmec-specific controlled research pilot.

AGENDA

Seven sections, one evidence-first storyline



01

Solution introduction

What was built, for whom, and within what safety boundary.



02

Problem assessment

Why fragmented records and ungrounded AI fail clinical review.



03

Business proposal

The doctor-centered workflow and where it creates value.



04

Technical proposal

Architecture, RAG pipeline and evaluation framework.



05

Delivery & demo readiness

Benchmark evidence, artifacts and a repeatable local demo.



06

Vinmec research / pilot bridge

How PoC evidence becomes a governed pilot design.



07

Limitations, maintenance, roadmap

Known boundaries, handover and conservative next steps.

Fluent text is not enough in clinical summarization



Fragmented records

A single encounter spans admission notes, progress notes, meds, allergies, labs, imaging and discharge planning across documents and sections.



Fluent but unsupported

Generic LLM output can sound confident without source support — risking summaries that omit or invent clinically important details.



Metrics miss grounding

ROUGE / BERTScore can reward lexical or semantic similarity while completely missing citation weakness or critical omissions.



*Clinicians need evidence traceability, scoped retrieval and visible uncertainty — **not just fluent prose.***

WHY THIS SOLUTION

RAG as evidence infrastructure, not a correctness guarantee



Evidence traceability

Important clinical claims can point back to the exact source evidence behind them.



Patient / encounter boundary

Retrieval is scoped to the selected patient and encounter — evidence cannot leak across records.



Safe refusal & gating

Missing mandatory evidence blocks or flags generation instead of forcing unsupported output.



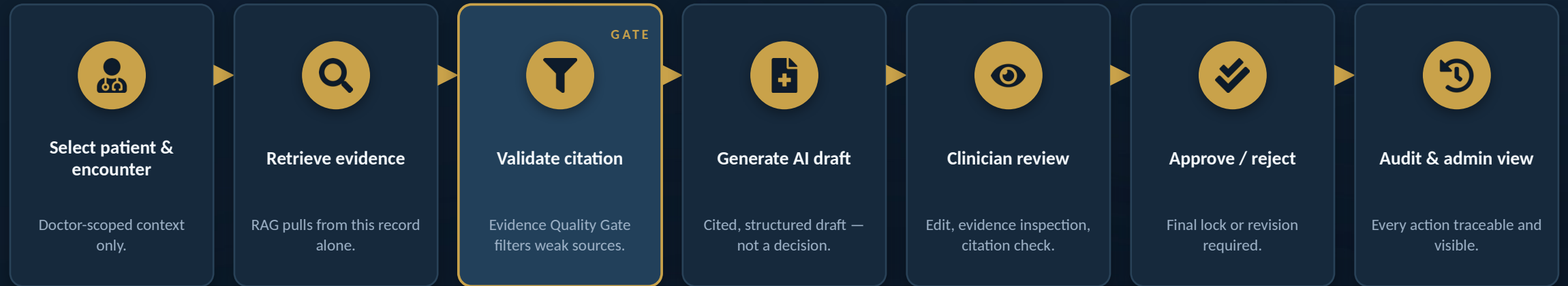
Reviewer efficiency

Clinicians inspect targeted evidence chunks rather than re-reading entire source documents.

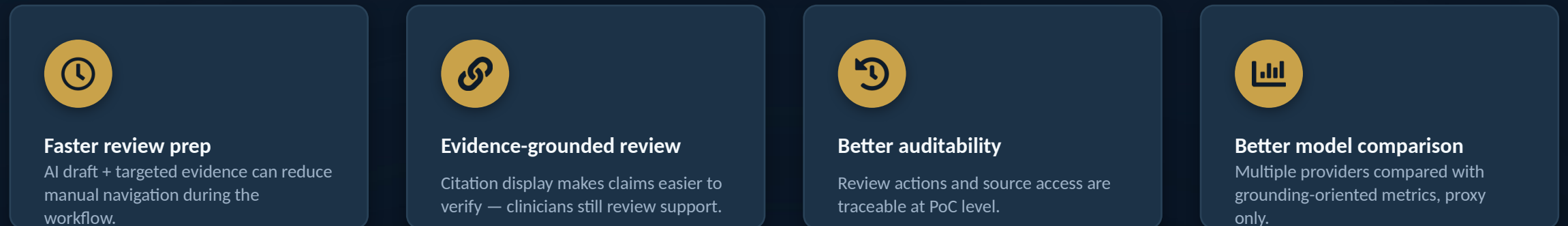
*“RAG helps structure evidence; **it does not guarantee clinical correctness.**”*

BUSINESS PROPOSAL

Solution at a glance — gated before any final use



BUSINESS VALUE



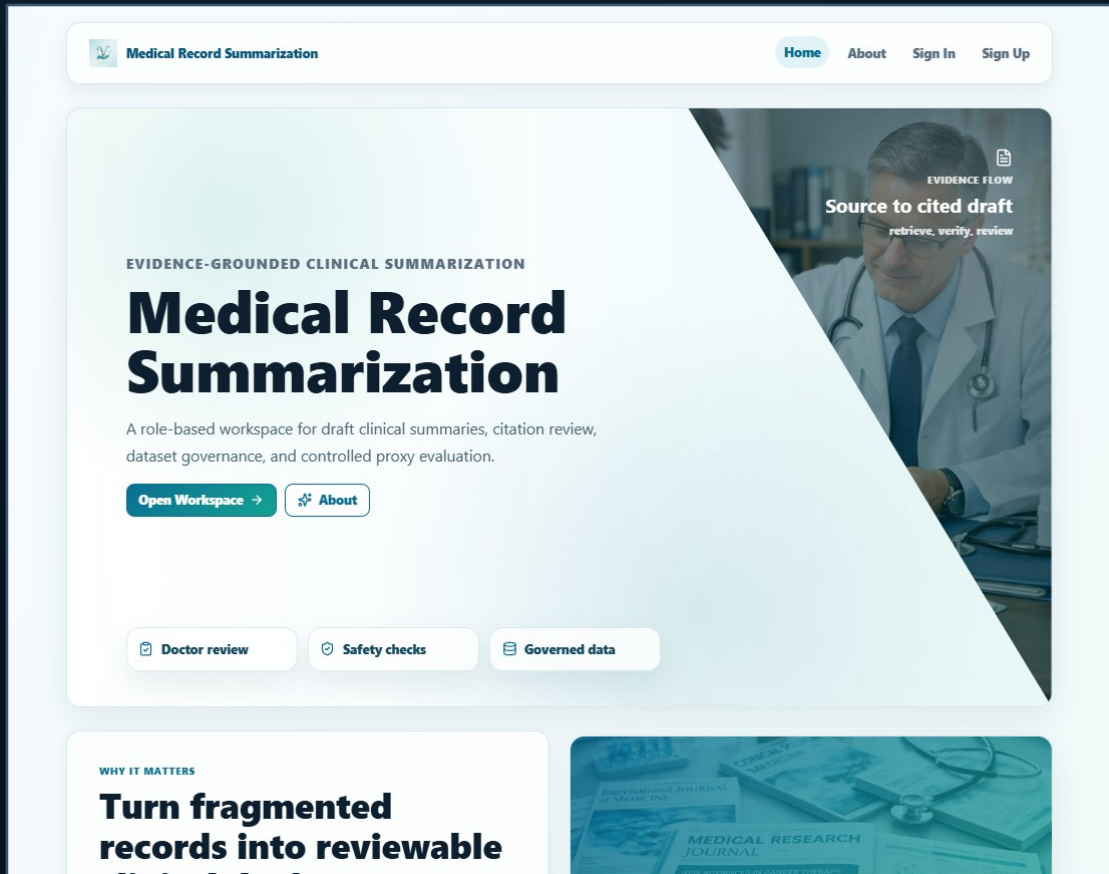
SECTION 03 • PRODUCT PROPOSAL

Inside the actual review workspace

Real screenshots from the running local PoC — not illustrations. Product entry through clinician decision and audit.

A role-based workspace, not a script

Real screenshot — local Docker Compose staging demo.

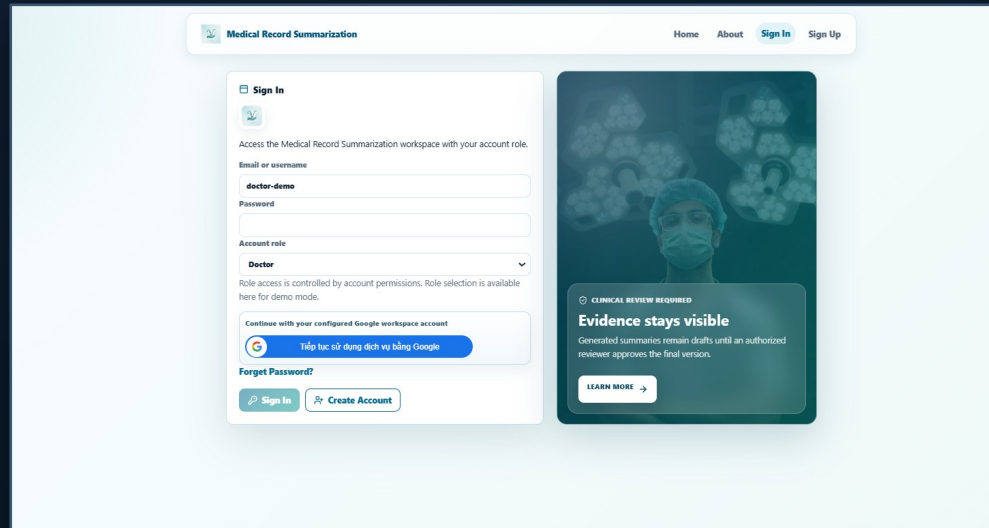


EVIDENCE-GROUNDED CLINICAL SUMMARIZATION

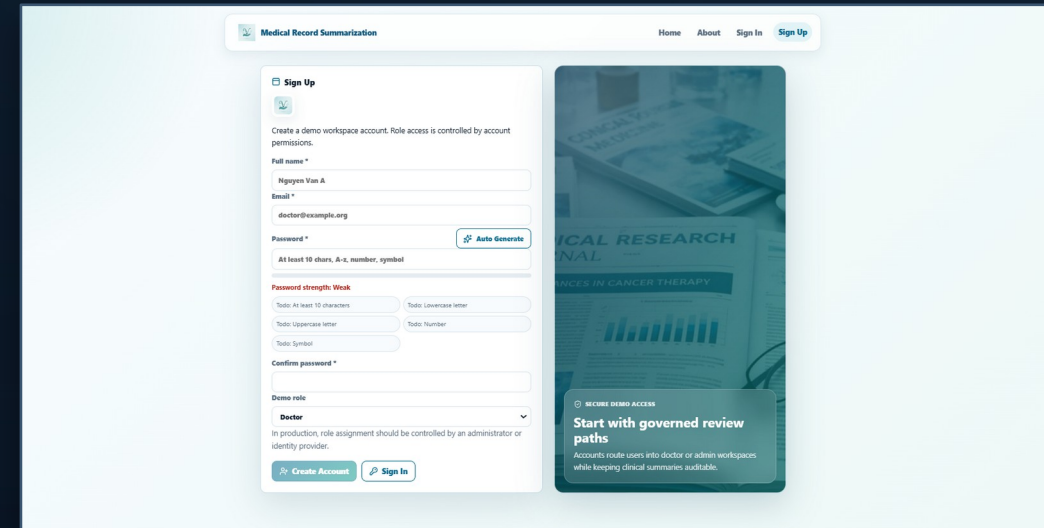
- Positioned explicitly as evidence-grounded, not a generic chatbot.
- Landing page names the three pillars up front: doctor review, safety checks, governed data.
- “Source to cited draft — retrieve, verify, review” sets the workflow expectation before login.

Governed, role-based account access

Real screenshots — role selection is explicit, and password policy is enforced at sign-up.



LOGIN • ROLE SELECTION

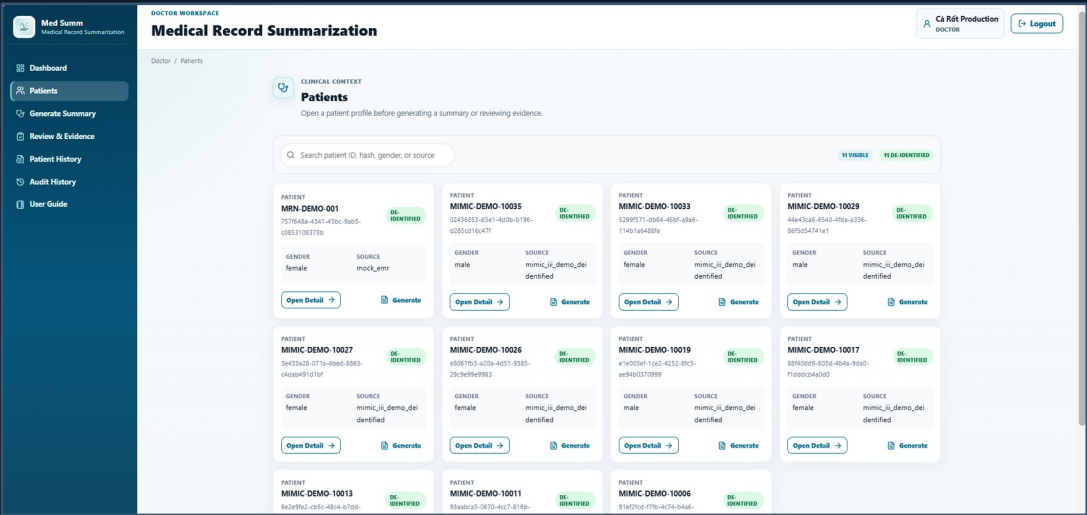


SIGN-UP • PASSWORD POLICY

Demo-mode note: role selection is exposed at login for demo access only — the product itself states that production role assignment should be controlled by an administrator or identity provider.

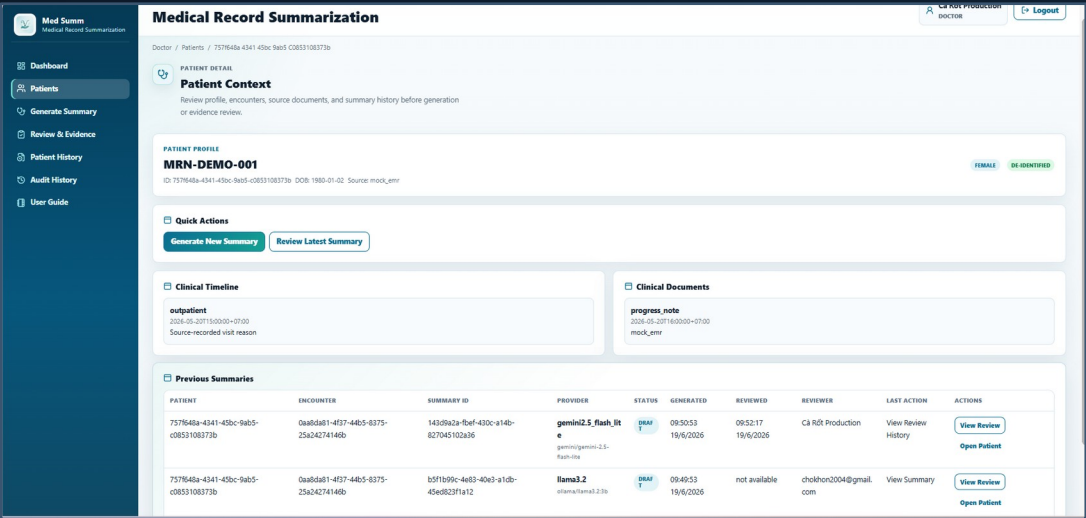
Patient & encounter scope

Scoping to a de-identified patient and encounter is the first safety control — real screenshots.



DE-IDENTIFIED PATIENT LIST

- Patient list shows de-identification status (MIMIC-III demo / mock EMR) directly on each card — 11/11 visible records flagged DE-IDENTIFIED.
- Patient Context combines profile, clinical timeline, source documents and prior summary history before any generation happens.



PATIENT CONTEXT & TIMELINE

Evidence-first draft generation

Real screenshot — patient-scoped MiniLM + Qdrant retrieval runs before provider inference.

Med Summ
Medical Record Summarization

Dashboard
Patients
Generate Summary
Review & Evidence
Patient History
Audit History
User Guide

DOCTOR GOLDEN PATH
Generate Summary
Create a draft patient snapshot. Evidence review, editing, and approval happen on the Review & Evidence page.

FLOW 2 / RAG EVIDENCE-FIRST

Doctor generation now uses patient-scoped MiniLM + Qdrant retrieval before provider inference.

Notes are chunked, embedded, retrieved by clinical section, checked by a quality gate, then sent to the selected provider as cited context.

Patient & Source

Patient search

Search patient ID, hash, or gender

MRN-DEMO-001 female **DE-ID**
MIMIC-DEMO-10035 male **DE-ID**
MIMIC-DEMO-10033 female **DE-ID**
MIMIC-DEMO-10029 male **DE-ID**
MIMIC-DEMO-10027 female **DE-ID**
MIMIC-DEMO-10026 female **DE-ID**
MIMIC-DEMO-10019 male **DE-ID**
MIMIC-DEMO-10017 female **DE-ID**
MIMIC-DEMO-10013 female **DE-ID**
MIMIC-DEMO-10011 female **DE-ID**

Selected patient: **MRN-DEMO-001** **DE-IDENTIFIED**
Encounter
outpatient
Source document
progress_note

Provider

Summary Provider

SELECTED Gemini 2.5 Flash Lite **READY** **Change**

Patient MRN-DEMO-001
Encounter outpatient
Type API provider
Context mode **RAG MINILM + QDRANT**

Draft only. Review evidence before approval.

PROVIDER READY
WORKER READY

Draft generated 125

JOB 34D8E3DD...D208 **DRAFT_READY**

Retrieval quality gate: warning. Missing required: none.

Patient scope
Looking patient and encounter context

Retrieve evidence
MiniLM + Qdrant section-aware evidence search

Build clinical context
Diagnosis, medications, timeline, diagnostics, assessment, plan

Provider generation
Gemini 2.5 Flash Lite cloud provider

Citation validation
Checking unsupported claims and missing evidence

Draft ready
Open Review & Evidence before approval

Generate Draft

Draft Preview **DRAFT**

Summary ID 16f5032a...bf6395
Provider gemini2.5_flash_lite
Generated 16:54:39 24/6/2026

Patient Snapshot

- Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]
- Age: 46 [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]
- Data status: de-identified [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

Diagnosis Evidence

- Condition: Source-recorded condition [chunk:e2c36bc8-a775-5d04-bdc6-3d62ff489aa5]
- Clinical status: active [chunk:e2c36bc8-a775-5d04-bdc6-3d62ff489aa5]

Medication Evidence

- Medication: Source-recorded medication [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]
- Status: active [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]
- Dosage: source-recorded dosage [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

Timeline Evidence

- Encounter type: outpatient [chunk:7cad3932-cff9-5]

Needs Clinician Review

- Provider output requires clinician review before clinical use.

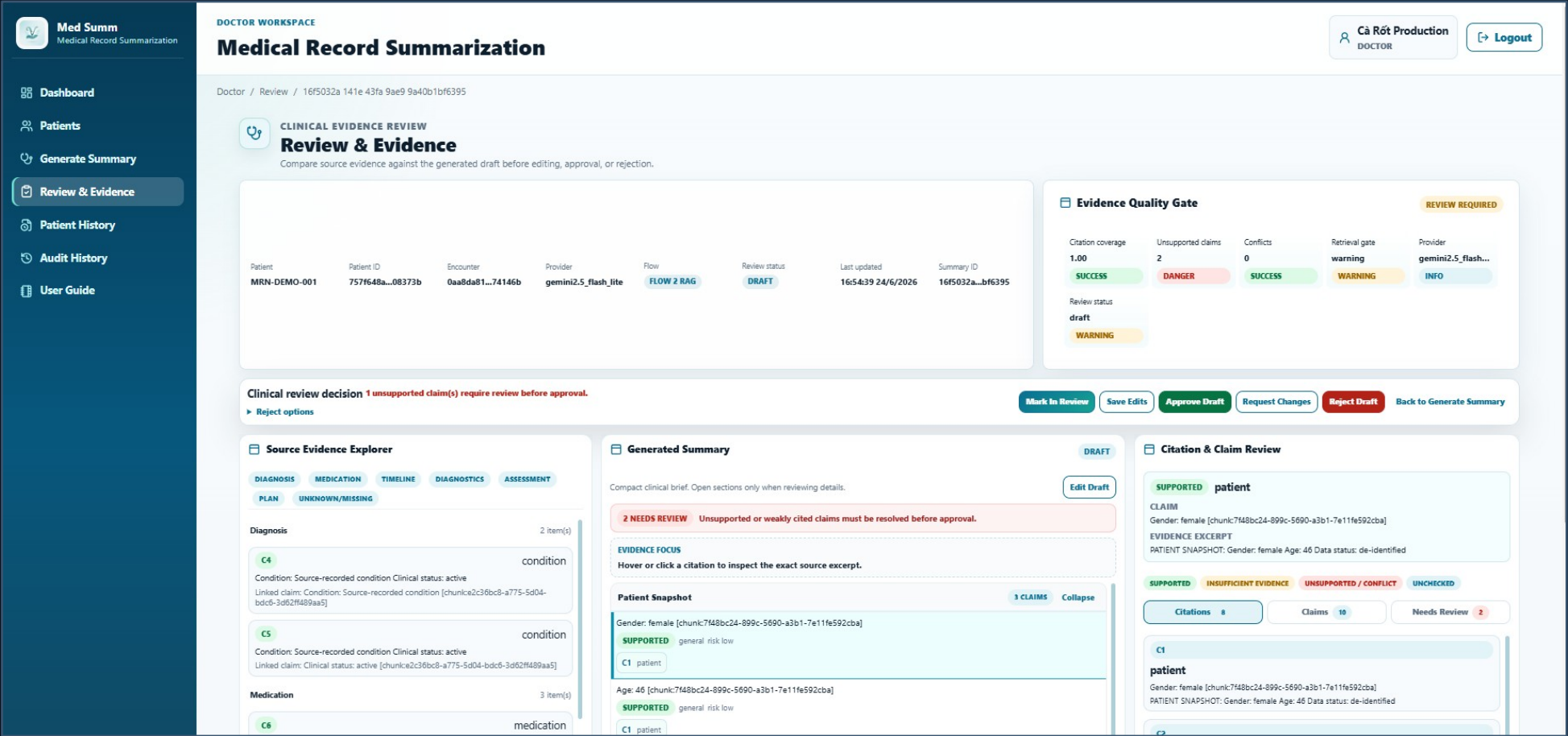
Review Evidence

Citation-grounded Medical Record Summarization PoC • Clinician-review-only • Proxy/de-identified data

11 / 30

Review & Evidence — the Evidence Quality Gate

Real screenshot — citation coverage, unsupported claims, conflicts and retrieval warnings, all visible before approval.



PRODUCT WALKTHROUGH • STEP 04

Citation is the review mechanism, not decoration

Real screenshots — every claim resolves to a source excerpt; hovering a citation shows the exact chunk.

The screenshot displays a clinical review decision interface. At the top, a status bar indicates "Clinical review decision 1 unsupported claim(s) require review before approval." and includes buttons for "Mark In Review", "Save Edits", "Approve Draft", "Request Changes", "Reject Draft", and "Back to Generate Summary".

The interface is divided into three main sections:

- Source Evidence Explorer:** Contains tabs for "DIAGNOSIS", "MEDICATION", "TIMELINE", "DIAGNOSTICS", and "ASSESSMENT". Under "MEDICATION", it shows a list of items with details like "Medication: Source-recorded medication Status: active Dosage: source-recorded dosage". Under "Unknown/Missing", it shows a list of items with details like "PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified".
- Generated Summary:** A compact clinical brief. It includes a "2 NEEDS REVIEW" alert: "Unsupported or weakly cited claims must be resolved before approval." It also has an "EVIDENCE FOCUS" section with a prompt: "Hover or click a citation to inspect the exact source excerpt." Below this, it shows a "Patient Snapshot" with details like "Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]" and "Age: 46 [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]". It also includes a "Diagnosis Evidence" section with details like "Condition: Source-recorded condition [chunk:2c35bc8-a775-5d04-bdc6-3d62f489aa5]".
- Citation & Claim Review:** Shows a list of citations and claims. It includes a "SUPPORTED" patient claim, an "INSUFFICIENT EVIDENCE" patient claim, and an "UNSUPPORTED / CONFLICT" patient claim. It also shows a "C1 patient" claim, a "C2 patient" claim, a "C3 patient" claim, and a "C4 condition" claim.

This screenshot shows a citation hover detail. It displays a list of citations with details like "C1 patient" and "C2 patient". It also shows a "Diagnosis Evidence" section with details like "Condition: Source-recorded condition [chunk:2c35bc8-a775-5d04-bdc6-3d62f489aa5]".

CITATION HOVER DETAIL — SOURCE EXCERPT ON DEMAND

PRODUCT WALKTHROUGH • STEP 05

The draft remains a draft until a clinician decides

Real screenshot — this summary was rejected; the system records the rejection and re-opens for a new draft.



Med Summ

Medical Record Summarization

Dashboard

Patients

Generate Summary

Review & Evidence

Patient History

Audit History

User Guide

Clinical review decision 1 unsupported claim(s) require review before approval. Summary rejected.

Reject options

Mark In ReviewSave EditsApprove DraftRequest ChangesReject DraftBack to Generate Summary

Summary rejected

Summary ID16f5032a-141e-43fa-9ae9-9a40b1bf6395PatientMRN-DEMO-001ReviewerCà Rôt ProductionTimestamp17:00:46 24/6/2026

The draft has moved to rejected status. You can edit the summary or generate a new draft.

View Patient HistoryOpen Audit HistoryGenerate New Draft

Source Evidence Explorer

DIAGNOSIS

MEDICATION

TIMELINE

DIAGNOSTICS

ASSESSMENT

PLAN

UNKNOWN/MISSING

Medication3 item(s)

dosage

Linked claim: Medication: Source-recorded medication [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

C7medication

Medication: Source-recorded medication Status: active Dosage: source-recorded dosage

Linked claim: Status: active [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

C8medication

Medication: Source-recorded medication Status: active Dosage: source-recorded dosage

Linked claim: Dosage: source-recorded dosage [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

Unknown/Missing3 item(s)

C1patient

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

Linked claim: Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

C2patient

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

Linked claim: Age: 46 [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

Generated Summary

REJECTED

Compact clinical brief. Open sections only when reviewing details.

Back to Review

2 NEEDS REVIEW Unsupported or weakly cited claims must be resolved before approval.

EVIDENCE FOCUS

Hover or click a citation to inspect the exact source excerpt.

Editable summary

Diagnosis Evidence

- Condition: Source-recorded condition [chunk:e2c36bc8-a775-5d04-bdc6-3d62ff489aa5]

- Clinical status: active [chunk:e2c36bc8-a775-5d04-bdc6-3d62ff489aa5]

Medication Evidence

- Medication: Source-recorded medication [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

- Status: active [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

- Dosage: source-recorded dosage [chunk:27174b40-52d8-5a69-80ce-5efdbc42bca9]

Timeline Evidence

Encounter type: outpatient [chunk:7e042022-4660-5...

Save Edit

Version 1 Citation coverage 1.0000 1 unsupported claims

Citation & Claim Review

SUPPORTED patient

CLAIM

Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

EVIDENCE EXCERPT

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

SUPPORTEDINSUFFICIENT EVIDENCEUNSUPPORTED / CONFLICTUNCHECKED

Citations8Claims10Needs Review2

Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

C2patient

Age: 46 [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

C3patient

Data status: de-identified [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

PATIENT SNAPSHOT: Gender: female Age: 46 Data status: de-identified

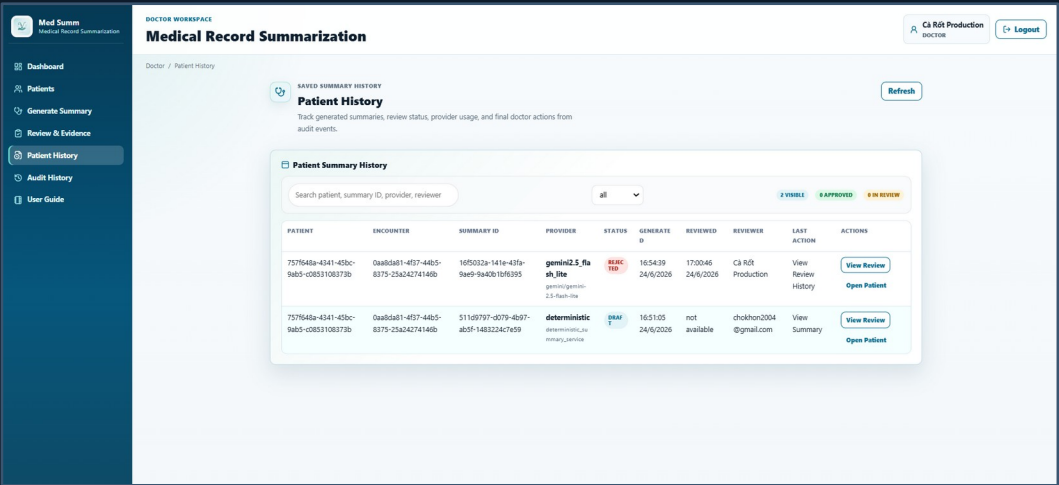
C4condition

Condition: Source-recorded condition [chunk:e2c36bc8-a775-5d04-bdc6-3d62ff489aa5]

Condition: Source-recorded condition Clinical status: active

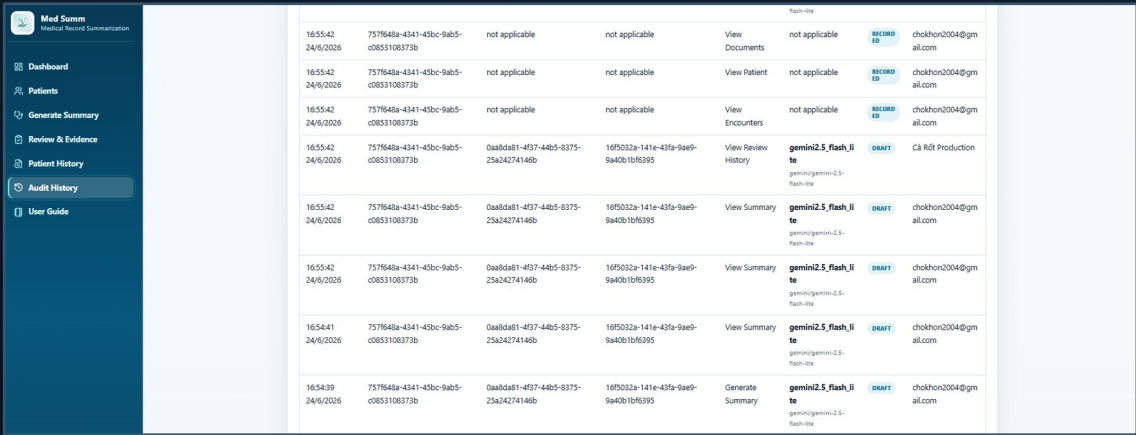
Lifecycle and auditability

Real screenshots — summary history and the underlying audit event log.



PATIENT SUMMARY HISTORY

- History tracks provider, status (REJECTED / DRAFT), generated time, reviewer and last action per summary version.
- The audit log is a flat, timestamped event trace — every view, generate, and review action is recorded with provider and reviewer identity.



RAW AUDIT EVENT LOG

Five stakeholders, five distinct kinds of value



Doctor / Clinician Reviewer

Fast draft with evidence
visibility — full human control
retained.

Patient/encounter selection ·
Review & Evidence workspace ·
final lock



Admin / Evaluator

One place to compare
providers and inspect
benchmark quality.

Flow 2.1 RAG Best Models
dashboard · audit & PHI-safe
export intent



Technical Reviewer

Confidence that the demo is
repeatable and lightweight to
run.

Docker Compose · /health ·
/ready · runtime dependency
separation



Research Evaluator

Evaluation that goes beyond
ROUGE/BERTScore alone.

Citation coverage · unsupported-
claim rate · factuality/omission
proxies



Project Mentor / Reviewer

A clear, conservatively-framed
project story and forward
path.

Delivery reports · this proposal ·
explicit limitations

TECHNICAL PROPOSAL

Separated layers — product workflow apart from evaluation & runtime



Frontend

React + Vite

Doctor workspace · Admin Evaluation UI · Review & Evidence surfaces



Backend / API

FastAPI + RBAC

Auth · Patient/Encounter APIs · Summary, Citation, Safety, Review, Eval services



AI / RAG layer

Retrieval & generation

Section-aware chunking · MiniLM/hashing embeddings · Qdrant-compatible store · provider gateway



Citation & safety

Grounding checks

Source-evidence view · wrong-patient prevention · coverage & conflict scoring



Data & runtime

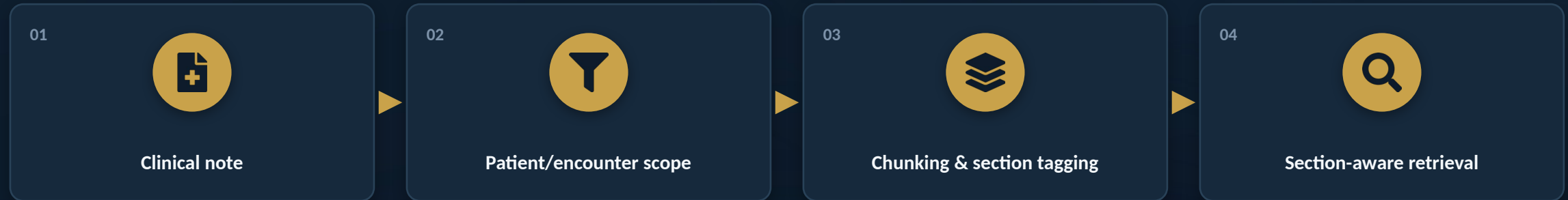
PostgreSQL · Redis

Docker Compose · RQ worker · /health · /ready · CSV/JSONL/report artifacts

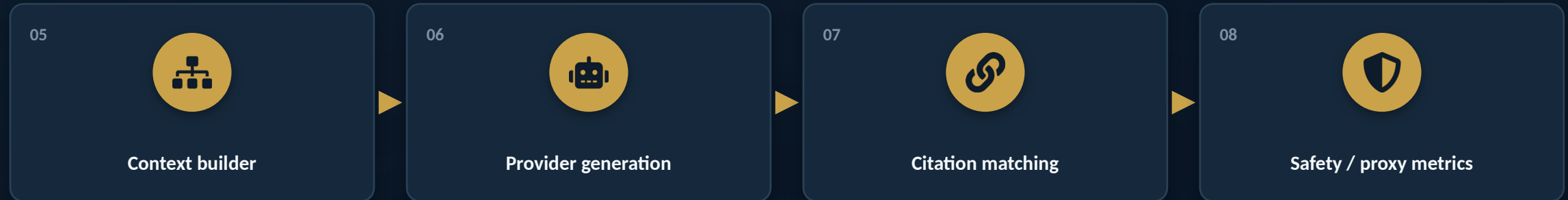
RAG & CITATION PIPELINE

Chunking precedes retrieval; retrieval supports citation review

STEPS 1-4 • EVIDENCE PREPARATION



STEPS 5-8 • GENERATION & SAFETY



Clinician review → **final lock & audit**. Qdrant-compatible retrieval finds candidate evidence — it does not replace review.

SECTION 05-06 • EVIDENCE & PILOT BRIDGE

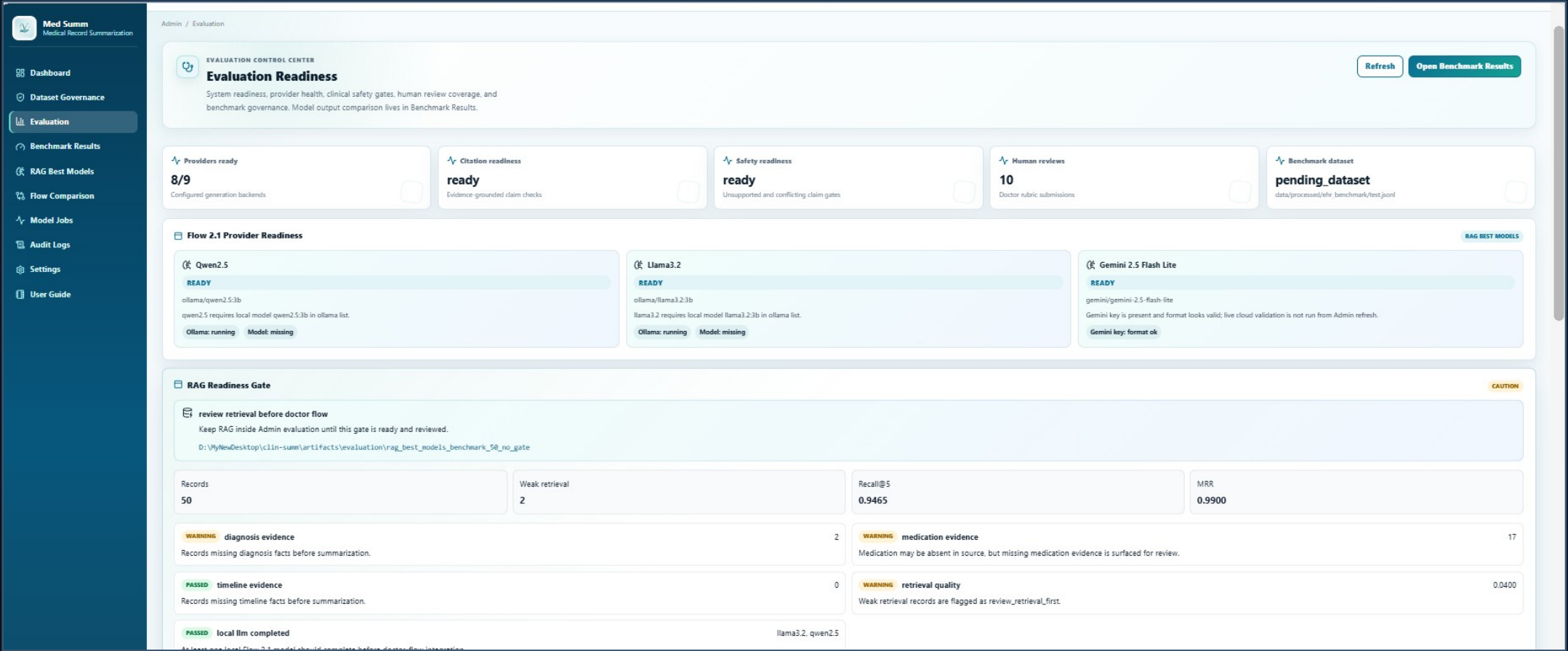
Admin evaluation evidence and the road to a pilot

Real benchmark screenshots, technical readiness proof, and a conservative path to a governed Vinmec research pilot.

ADMIN EVALUATION VISIBILITY

Evaluation discipline is visible, not hidden in scripts

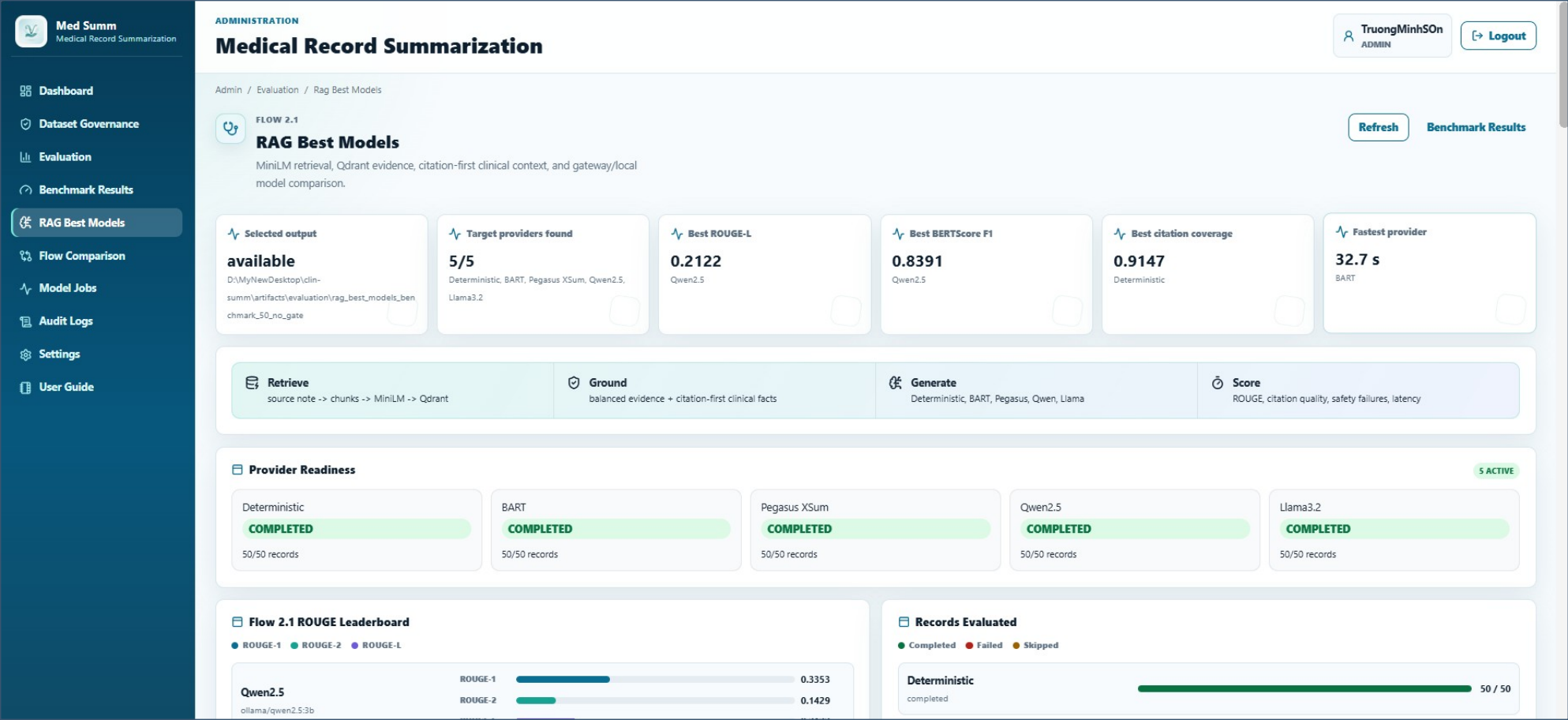
Real screenshot — provider readiness, citation/safety readiness, and the RAG retrieval-quality gate.



BENCHMARK RESULT SUMMARY

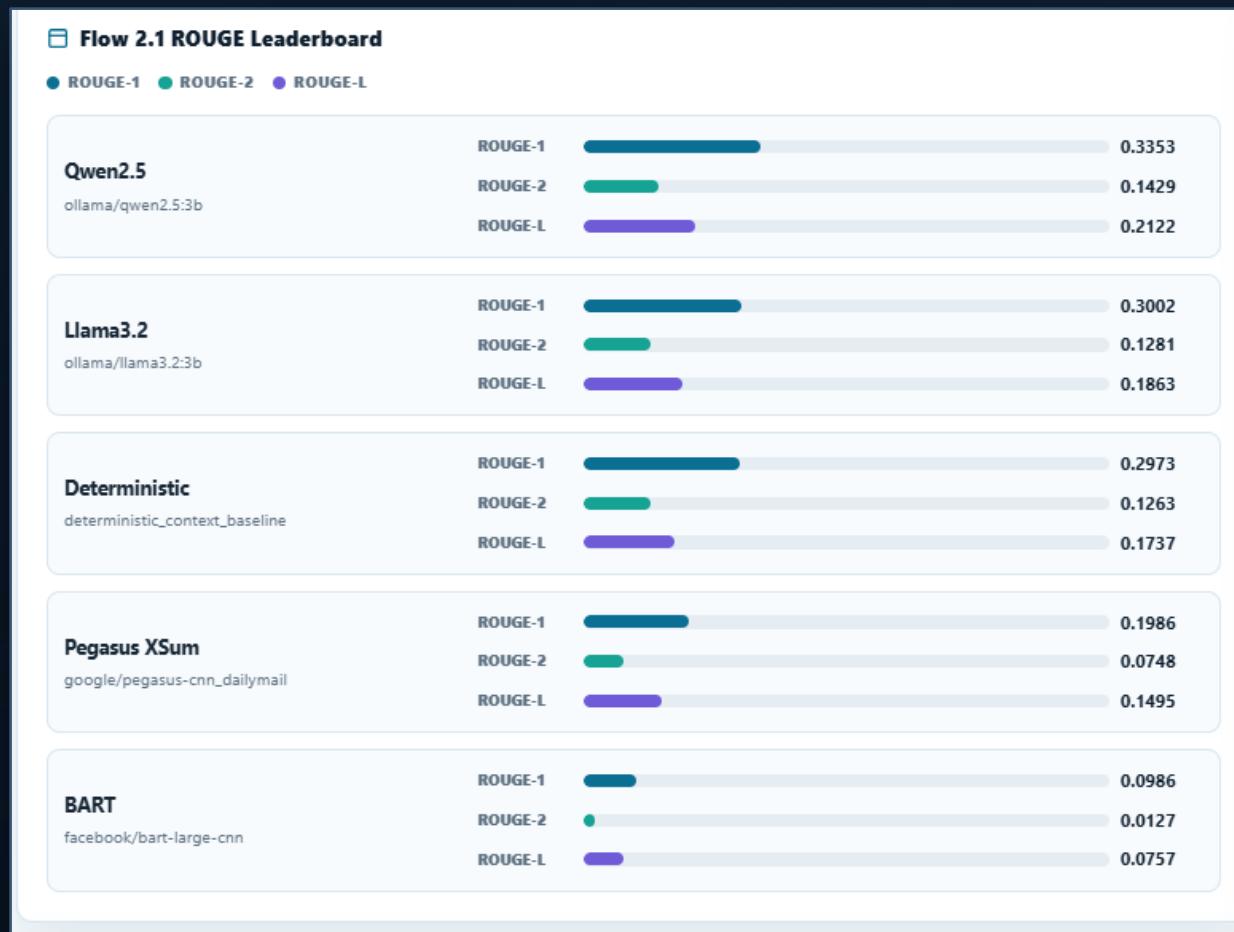
RAG Best Models — Flow 2.1 benchmark overview

Real screenshot — the exact admin screen the proposal's benchmark numbers are sourced from.



Flow 2.1 ROUGE leaderboard

Real screenshot — ROUGE-1 / ROUGE-2 / ROUGE-L across all five providers.



EVIDENCE GROUNDING

Citation coverage, faithfulness and omission — by provider

Real screenshot — the exact per-provider grounding table behind every number in this proposal.

Evidence Grounding

Best faithfulness proxy

PER_RECORD_METRICS.CSV

Deterministic

Deterministic	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
	0.9147	0.0000	0.9071	0.3688
deterministic_context_baseline	Timeline completeness			
	0.6667			

BART

BART	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
	0.1307	0.0000	0.7585	0.9583
facebook/bart-large-cnn	Timeline completeness			
	0.0645			

Pegasus XSum

Pegasus XSum	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
	0.3800	0.0100	0.7626	0.9236
google/pegasus-cnn_dailymail	Timeline completeness			
	0.1290			

Qwen2.5

Qwen2.5	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
	0.8884	0.0346	0.8713	0.4460
ollama/qwen2.5:3b	Timeline completeness			
	0.4785			

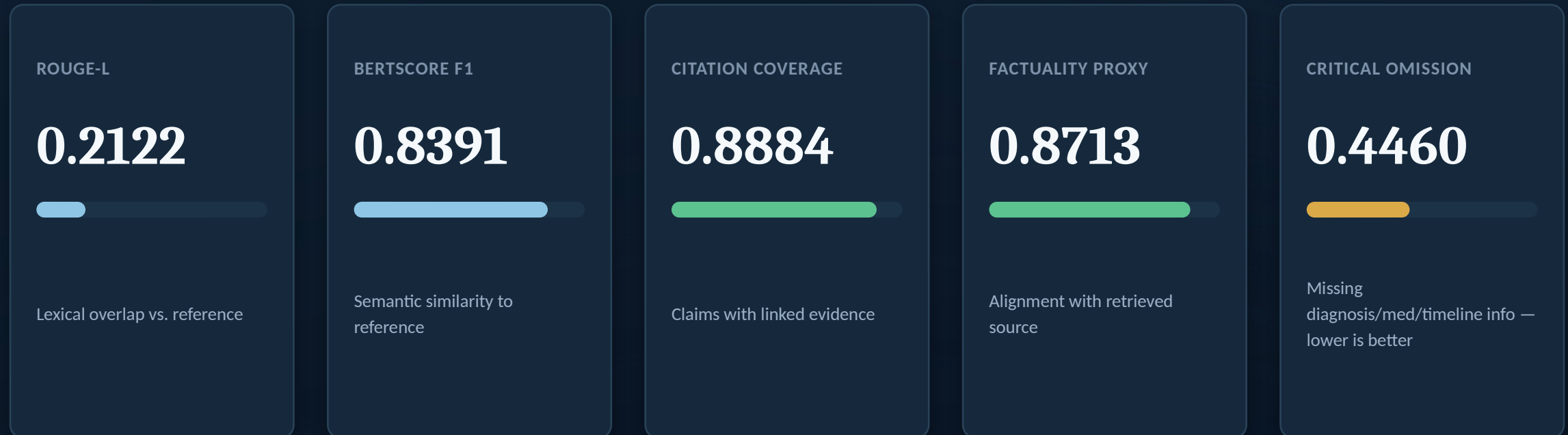
Llama3.2

Llama3.2	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
	0.8620	0.0618	0.8413	0.5108
ollama/llama3.2:3b	Timeline completeness			
	0.4570			

- Deterministic: 0.9147 citation coverage, 0.0000 unsupported — the stable control.
- Qwen2.5: 0.8884 coverage, 0.8713 faithfulness — strongest generative provider.
- Llama3.2: highest critical-omission rate among generative models at 0.5108.
- BART / Pegasus: weak citation coverage (0.13 / 0.38) — poor fit for citation-first review.

The strongest balanced provider in the completed proxy run

Use these numbers as evidence for pilot prioritization — not as proof of clinical safety.



Note: Qwen2.5 also shows a hallucination proxy of 0.48 — stronger grounding does not eliminate the need for clinician review.

WHY RAG HELPS

Same record, same model — raw vs. context vs. RAG-grounded

Real screenshot — retrieval-grounded generation reduces missing diagnosis, medication and timeline gaps.

Med Summ
Medical Record Summarization

Dashboard

Dataset Governance

Evaluation

Benchmark Results

RAG Best Models

Flow Comparison

Model Jobs

Audit Logs

Settings

User Guide

Compare raw summarization, clinical context, and RAG-grounded generation on the same record and same model. Use this page to explain when retrieval improves clinical completeness and when evidence retrieval needs review.

Three-Flow Same Record Comparison

All models

Refresh

Compare the same record and same model across raw summarization, sectioned clinical context, and RAG-grounded generation. This panel is meant to answer: **why do we need RAG?**

Proxy evaluation only. These results do not demonstrate clinical safety, clinical effectiveness, or real-world healthcare performance. Real EHR evaluation requires credentialed datasets such as MIMIC-IV-Note or MIMIC-IV-BHC under approved governance processes.

Deterministic
multiclinsum_ls_en_10015

RAG HELPED

Deterministic
multiclinsum_ls_en_10033

RAG HELPED

BART
multiclinsum_ls_en_10033

RAG HELPED

Pegasus XSum
multiclinsum_ls_en_10002

RAG HELPED

Pegasus PubMed
multiclinsum_ls_en_10021

RAG HELPED

Deterministic
multiclinsum_ls_en_10002

RAG HELPED

Pegasus PubMed

Same record, same model
multiclinsum_ls_en_10015 - Deterministic

Missing diagnosis
+0.2500

Missing medication
+1.0000

Timeline completeness
+0.3334

Unsupported claims
0.0000

Citation coverage
+0.8571

Flow 1 Rawdeterministic_sentence_baselineCOMPLETED

ROUGE-L
0.1317

Diagnosis miss
1.0000

Medication miss
1.0000

Timeline
0.3333

Unsupported
0.0000

Citation
0.0000

MISSING DIAGNOSIS

MISSING MEDICATION

MISSING TIMELINE

INCOMPLETE SUMMARY

RETRIEVAL-RELATED FAILURE

SOURCE DATA LIMITATION

A 78-year-old Israeli man presented to our intensive care unit with fever, flaccid limb weakness, and dysarthria. On the morning of his admission he felt cold and weak. He awoke suddenly with vomiting, weakness of four limbs, and slurred speech.

Flow 1.5 Contextdeterministic_context_baselineCOMPLETED

ROUGE-L
0.0331

Diagnosis miss
1.0000

Medication miss
1.0000

Timeline
0.0000

Unsupported
0.0000

Citation
0.8571

HALLUCINATED CONTENT

MISSING DIAGNOSIS

MISSING MEDICATION

MISSING TIMELINE

INCOMPLETE SUMMARY

RETRIEVAL-RELATED FAILURE

Use only the evidence below. Preserve diagnoses, medications, timeline, assessment, and plan. Do not add clinical facts that are not supported by evidence. [DIAGNOSIS]
- not available in retrieved evidence

[MEDICATIONS]
- (54af0def-7ea1-5af8-8300-38fc53ffdc4c) He had non-oliguric renal dysfunction.

Flow 2 RAGdeterministic_context_baselineCOMPLETED

ROUGE-L
0.1442

Diagnosis miss
0.7500

Medication miss
0.0000

Timeline
0.6667

Unsupported
0.0000

Citation
0.8571

MISSING DIAGNOSIS

(96c5e50b-5690-5bc1-9773-9efa2f7af89c) He was a retired lifeguard, and his past medical history included biologic aortic valve replacement 3 years earlier because of aortic stenosis, paroxysmal atrial fibrillation treated with apixaban anticoagulant therapy, an episode of atrial flutter treated with ablation, status p (885cec10-3dd5-5a67-9cfe-654046957da6) With regards to infectious diseases they included subacute prosthetic bacterial endocarditis with an embolic stroke, an infectious encephalitis (including herpes viruses, West Nile virus, sandfly encephalitis, Listeria monocytogenes, chlamydia, atypical bacterial infection (72a2b7f6-a222-552a- Build the summary from the cited clinical facts first. Use only the evidence below. Preserve diagnoses, medications, timeline, assessment, and plan. Do

Citation-grounded Medical Record Summarization PoC · Clinician-review-only · Proxy/de-identified data

25 / 30

REPRODUCIBILITY

Benchmark artifacts and run files — inspectable, not summarized away

Real screenshot — prediction files, run manifests and metric CSVs for every provider.

Med Summ
Medical Record Summarization

Dashboard

Dataset Governance

Evaluation

Benchmark Results

RAG Best Models

Flow Comparison

Model Jobs

Audit Logs

Settings

User Guide

Prediction Files

Deterministic

FOUND

deterministic_predictions.jsonl

50 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\deterministic_predictions.jsonl

BART

FOUND

bart_predictions.jsonl

50 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\bart_predictions.jsonl

Pegasus XSum

FOUND

pegasus_predictions.jsonl

50 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\pegasus_predictions.jsonl

Pegasus PubMed

MISSING

pegasus_pubmed_predictions.jsonl

0 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\pegasus_pubmed_predictions.jsonl

Pegasus CNN

MISSING

pegasus_cnn_dailymail_predictions.jsonl

0 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\pegasus_cnn_dailymail_predictions.jsonl

Qwen2.5

FOUND

qwen2.5_predictions.jsonl

50 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\qwen2.5_predictions.jsonl

Llama3.2

FOUND

llama3.2_predictions.jsonl

50 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\llama3.2_predictions.jsonl

Gemini 2.5 Flash Lite

MISSING

gemini2.5_flash_lite_predictions.jsonl

0 records

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\gemini2.5_flash_lite_predictions.jsonl

Run Artifacts

MODEL COMPARISON

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\model_comparison.csv

PER RECORD METRICS

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\per_record_metrics.csv

EVALUATION RUN MANIFEST

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\rag_benchmark_manifest.json

RAG BENCHMARK MANIFEST

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\rag_benchmark_manifest.json

RETRIEVAL METRICS

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\retrieval_metrics.csv

RETRIEVED EVIDENCE

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\retrieved_evidence.jsonl

RUN LOG

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\run.log

EVALUATION REPORT

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\EVALUATION_REPORT.md

PER RECORD FAILURE ANALYSIS

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\per_record_failure_analysis.jsonl

REPRODUCIBILITY MANIFEST

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\reproducibility_manifest.json

CITATION GROUNDING REPORT

D:\MyNewDesktop\clin-sum\artifacts\evaluation\rag_best_models_benchmark_50_no_gate\citation_grounding_report.md

PROVIDER	CHECKPOINT	STAGE	DOMAIN FIT	STATUS	RECORDS	ROUGE-1	ROUGE-2	ROUGE-L	BERTSCORE	FAITHFULNESS	CITATION	UNSUPPORTED	AVG LATENCY	P95 LATENCY
Deterministic	deterministic_context_baseline	not available	Fast extractive baseline	COMPLETED	50/50	0.2973	0.1263	0.1737	0.7952 computed roberta-large	0.9071	0.9147	0.0000	0 ms	0 ms
BART	facebook/bart-large-cnn	not available	General summarization baseline	COMPLETED	50/50	0.0986	0.0127	0.0757	0.8010 computed roberta-large	0.7585	0.1307	0.0000	32.7 s	59.0 s
Pegasus XSum	google/pegasus-cnn_dailymail	not available	General Pegasus baseline	COMPLETED	50/50	0.1986	0.0748	0.1495	0.8232 computed	0.7626	0.3800	0.0100	37.2 s	50.7 s

Citation-grounded Medical Record Summarization PoC · Clinician-review-only · Proxy/de-identified data

26 / 30

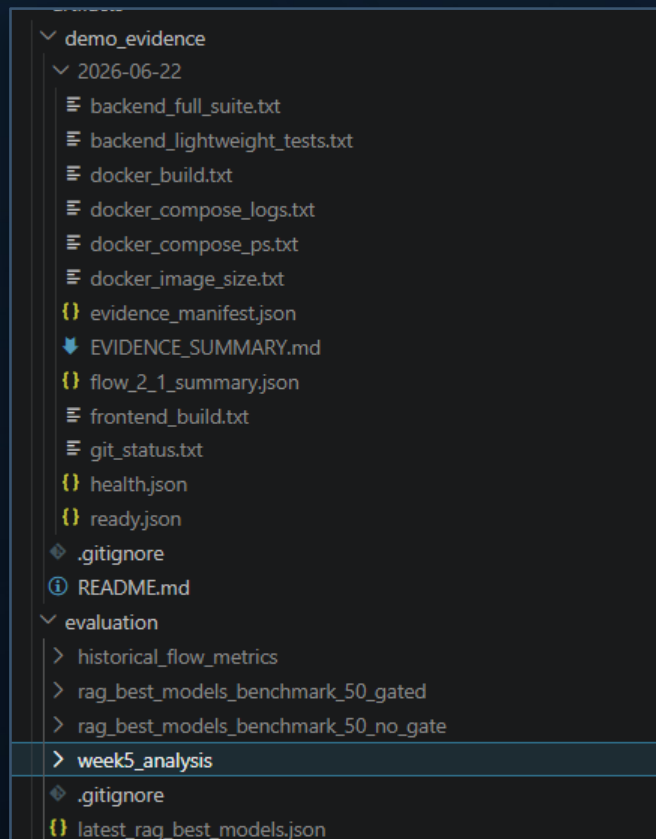
TECHNICAL READINESS

Local staging readiness, not production deployment

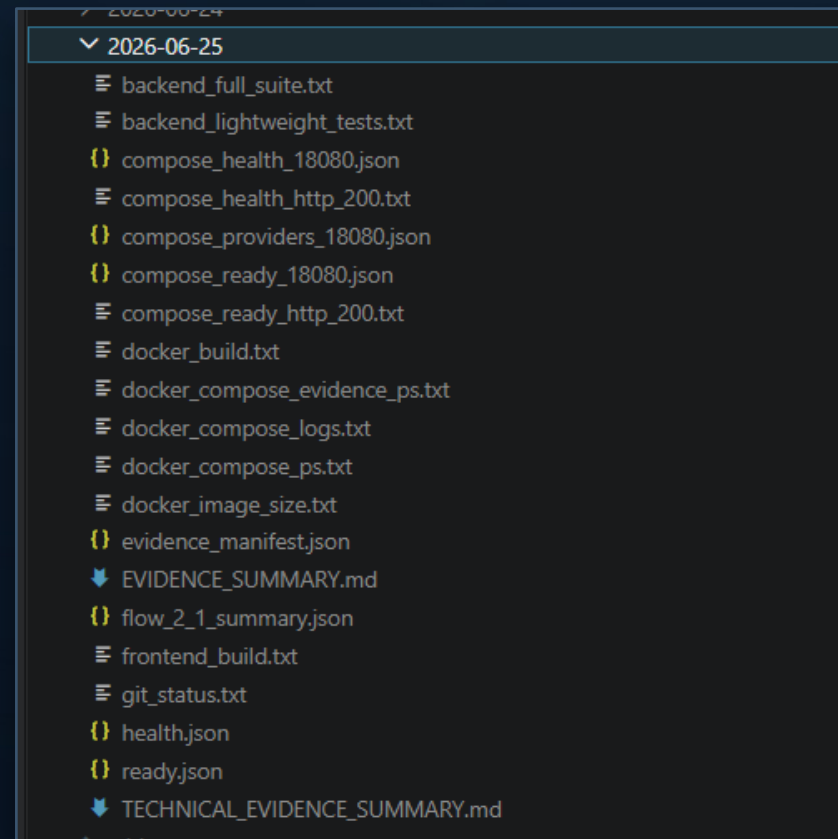
Real screenshots — the running-system checklist and the evidence-package folder structure.

Bảng chứng kỹ thuật	Command terminal cần chụp	Kết quả cần thấy
Docker Compose services running	<code>docker compose ps</code>	app , db , redis running/healthy
Health endpoint reachable	<code>Invoke-WebRequest http://127.0.0.1:8080/health Select-Object StatusCode,Content</code>	StatusCode: 200
Readiness endpoint reachable	<code>Invoke-WebRequest http://127.0.0.1:8080/ready Select-Object StatusCode,Content</code>	StatusCode: 200
Backend tests pass	<code>python -m pytest backend\tests -p no:cacheprovider -q</code>	passed
Frontend build pass	<code>cd frontend rồi npm run build</code>	built in ... hoặc build success
Docker build pass	<code>docker build -t clin-summ- local:demo .</code>	Successfully built / image created

RUNNING CHECKLIST (VN)

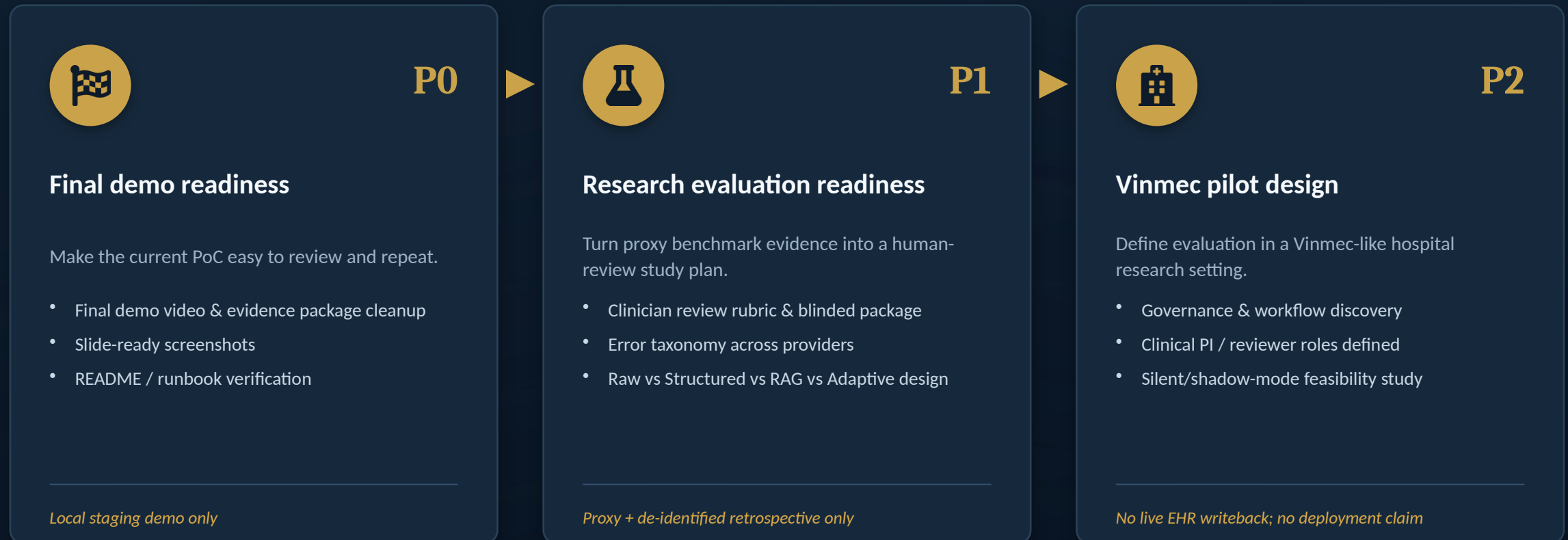


EVIDENCE PACKAGE • 2026-06-22



LATEST RUN • 2026-06-25

A staged move to governed research — not a hospital rollout



LIMITATIONS & RISK CONTROLS

The proposal is credible because it controls its risks

RISKS

- ✗ Wrong-patient / wrong-encounter evidence
- ✗ Unsupported or fabricated claims
- ✗ Medication or allergy detail errors
- ✗ PHI exposure in artifacts or logs
- ✗ Clinician over-trust in AI drafts

CONTROLS

- ✓ Patient/encounter scope filters & citation source checks
- ✓ Citation validation & unsupported-claim visibility
- ✓ Clinician edit/approve/reject before any final lock
- ✓ De-identified/mock data; PHI-safe audit export intent
- ✓ Mandatory human review; draft-only positioning throughout

CONCLUSION

A credible PoC — and a conservative path forward

- The product workflow shows how doctors can review an AI draft with evidence and auditability.
- The technical architecture shows how retrieval, citation, evaluation and local staging are organized.
- The benchmark shows proxy evidence and failure modes — not clinical safety or effectiveness.
- The Vinmec path proposes governed, de-identified, clinician-reviewed research — not deployment.

The final ask: approval to design a controlled pilot protocol — not to use the system clinically.